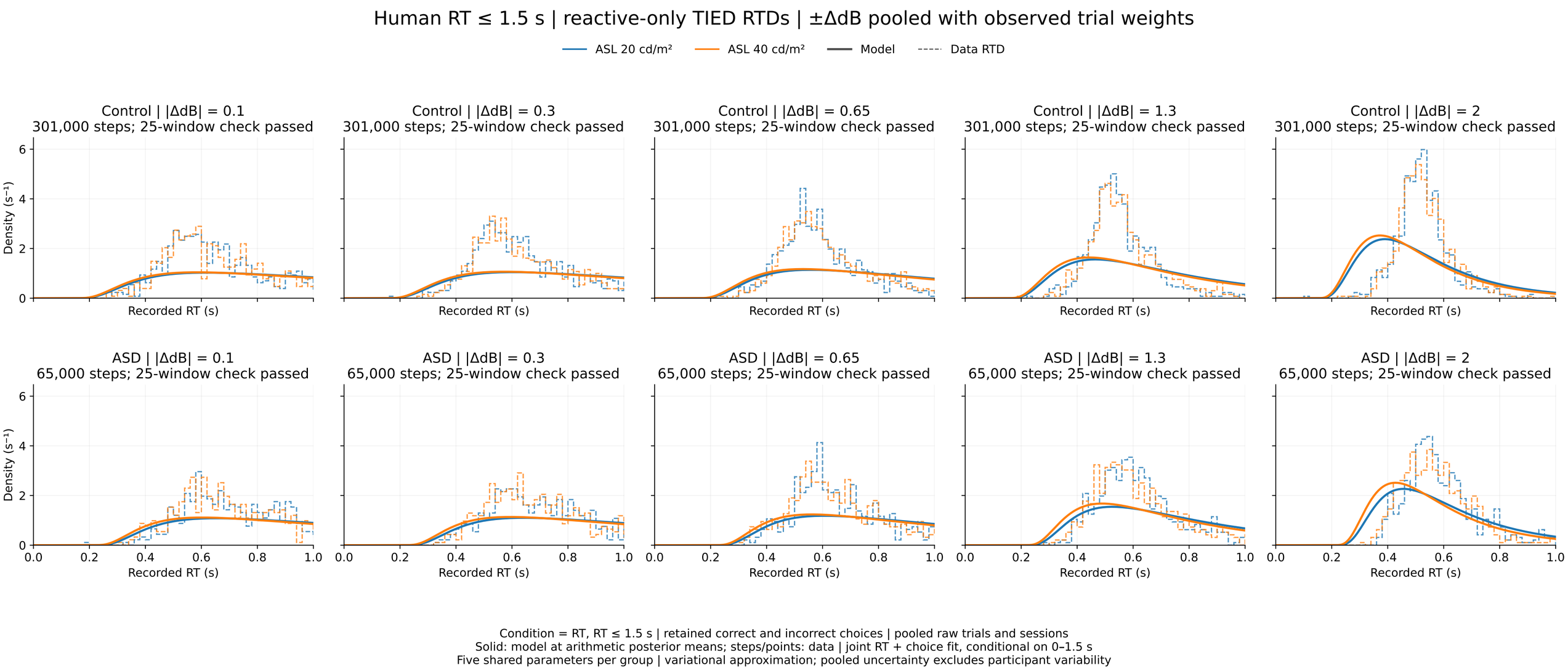
Human RT ≤ 1.5 s | reactive-only TIED RTDs | ±ΔdB pooled with observed trial weights
ASL 20 cd/m²
ASL 40 cd/m²
Model
Data RTD
Control | |ΔdB| = 0.1
301,000 steps; 25-window check passed
Control | |ΔdB| = 0.3
301,000 steps; 25-window check passed
Control | |ΔdB| = 0.65
301,000 steps; 25-window check passed
Control | |ΔdB| = 1.3
301,000 steps; 25-window check passed
Control | |ΔdB| = 2
301,000 steps; 25-window check passed
ASD | |ΔdB| = 0.1
65,000 steps; 25-window check passed
ASD | |ΔdB| = 0.3
65,000 steps; 25-window check passed
ASD | |ΔdB| = 0.65
65,000 steps; 25-window check passed
ASD | |ΔdB| = 1.3
65,000 steps; 25-window check passed
ASD | |ΔdB| = 2
65,000 steps; 25-window check passed
Density (s⁻¹)
Recorded RT (s)
0
2
4
6
0.0
0.2
0.4
0.6
0.8
1.0
Condition = RT, RT ≤ 1.5 s | retained correct and incorrect choices | pooled raw trials and sessions
Solid: model at arithmetic posterior means; steps/points: data | joint RT + choice fit, conditional on 0–1.5 s
Five shared parameters per group | variational approximation; pooled uncertainty excludes participant variability